Ali Feteha

AI & Software Engineer

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LinkedIn | GitHub | Portfolio

Professional Profile

AI & Software Engineer specializing in enterprise Generative AI, multi-agent systems, RAG, and real-time conversational AI. Delivered 5+ AI projects into production, with hands-on experience in agent orchestration, backend development, cloud integration, and automated evaluations. Experienced in leading technical delivery, collaborating across teams, and solving complex engineering challenges.

Technical Skills

AI & Machine Learning: Generative AI, Agentic AI, Multi-Agent Systems, Large Language Models (LLMs), RAG, Tool Calling, LLM Evaluation, NLP, Computer Vision

Frameworks & Libraries: Microsoft Agent Framework, Semantic Kernel, LangChain, FastAPI, PyTorch, TensorFlow, Scikit-learn

Azure & Cloud: Azure OpenAI, Azure AI Search, Microsoft Foundry, Azure AI Speech, Azure Document Intelligence, Azure Functions, Azure App Service, Azure Container Apps, Azure Cosmos DB

Programming & Backend: Python, SQL, REST APIs, Data Pipelines, API Integration, WebRTC, WebSockets

Development Tools: Git, Docker, Azure DevOps

Professional Experience

Applab | AI & Software Engineer Apr 2025 – Present

Doha, Qatar

- Delivered 5+ AI projects into production, including conversational AI platforms, multi-agent systems, RAG applications, and AI-driven workflow automation.
- Conduct R&D to investigate complex technical challenges, prototype multiple approaches, and develop solutions to key problems across AI projects.
- Partner with business teams in the early stages of projects to gather requirements, define AI use cases, estimate delivery timelines and team effort, and assess cloud architecture to ensure the necessary resources are in place.
- Provide technical guidance to junior and associate developers by reviewing proposed solutions, reviewing their code for correct implementation, and helping them resolve technical issues.
- Develop AI proofs of concept (POCs) to demonstrate technical feasibility and support potential client projects.

Accountant & Technical Office Engineer May 2020 – Feb 2025

Hassan Aly Hassan Association for Contracting | Alexandria, Egypt

- Managed finance-related tasks and supported technical-office operations, including quantity analysis across construction projects.

AI Projects Delivered

Project headings follow the format: Client | Project | Employer.

Qatar Olympic Committee (QOC) | Multi-Avatar AI Platform | Applab

LangChain · Azure OpenAI · Azure AI Search · Microsoft Foundry · Azure AI Speech · WebRTC

- Developed the initial proof of concept (POC), which was approved by QOC, and subsequently designed and implemented the production AI backend, including REST APIs, real-time avatar communication, and session management.
- Designed AI agent workflows using LangChain and Azure OpenAI, supporting dynamic avatar selection, Azure AI Search retrieval, and grounded web search through Microsoft Foundry. Developed website indexing pipelines and automated evaluations to assess agent performance.
- Implemented real-time dual-avatar session management and switching using WebRTC and WebSockets, collaborating with the frontend team to integrate AI-driven avatar transitions based on user queries.

Qatar Civil Aviation Authority (CAA) | Public AI Chatbot | Applab

Microsoft Semantic Kernel · Azure OpenAI · Azure AI Search · Azure Document Intelligence

- Designed and developed a multi-agent AI assistant using Microsoft Semantic Kernel, with specialized agents for aviation regulations, general inquiries, and service workflows.
- Engineered amendment-aware legal retrieval to identify the latest applicable aviation regulations and provide grounded responses with document citations and PDF page references.
- Developed automated content ingestion and indexing pipelines with incremental synchronization, document

extraction, OCR, and Arabic translation to support bilingual knowledge retrieval.

Qatar Olympic Committee (QOC) | Ertibat | Applab

Microsoft Agent Framework · Azure OpenAI · Workflow Automation

- Designed and developed an AI automation agent using Microsoft Agent Framework, integrating 50+ tools with backend APIs to execute business operations and automate workflows within the Ertibat platform.
- Implemented request preprocessing and API response transformations to handle different backend requirements and provide structured data for reliable agent tool execution.
- Built an end-to-end evaluation framework to validate agent tool calls and verify that executed operations were correctly reflected in the backend system.

Qatar Foundation (QF) | BOTaina / Maktabi | Applab

Microsoft Copilot Studio · Microsoft Agent Framework · Power Apps · SharePoint

- Took ownership of an existing enterprise AI assistant before production launch, resolving critical integration issues and contributing to production readiness across Copilot Studio and AI agent components.
- Designed and implemented permission-aware retrieval workflows using Microsoft Graph API, SharePoint REST API, and Power Automate, enabling the AI assistant to retrieve content based on individual user access permissions.
- Enhanced Azure Functions-based SharePoint content ingestion and indexing pipelines to support large volumes of enterprise data, improving content processing and retrieval across QF-selected sites.

Ministry of Justice (MOJ) | Public AI Assistant | Applab

Microsoft Agent Framework · Azure OpenAI · Azure AI Search · Azure AI Speech

- Led the AI engineering delivery of a public-facing AI assistant with chatbot and avatar capabilities, contributing to solution design, reviewing implementations, and providing technical guidance to developers.
- Collaborated with MOJ stakeholders and cross-functional teams to align technical requirements, integration approaches, and implementation activities.
- Managed AI backend deployments, handled bug fixes and enhancements, and resolved technical issues to support the successful public launch.

Research & Independent Projects

AHDoc | Arabic Handwritten Document OCR *Sep 2024 – Feb 2025*

PyTorch · Swin Transformer V2 · BERT · ResNet-50 · Computer Vision

- Developed an end-to-end Arabic handwritten document OCR system using a Swin Transformer V2–BERT architecture, achieving a 96%+ character recognition rate through transfer learning and data augmentation.
- Fine-tuned a ResNet-50-based text detection model using Differentiable Binarization and Adaptive Scale Fusion, achieving 81.66% precision and 79.0% F-measure.
- Developed automated data generation and augmentation pipelines to address training data limitations and OCR sequence-length challenges.

Education & Publication

Egypt-Japan University of Science and Technology (EJUST) Feb 2025

B.Sc. Computer Science and Engineering | Grade: Very Good | Alexandria, Egypt

IEEE NILES 2024: Waly, A., Tarek, B., Feteha, A., Yehia, R., Amr, G., & Fares, A. (2024). *Arabic handwritten document OCR solution with binarization and adaptive scale fusion detection.* View Publication.

Certifications & Training

Microsoft Certified: Azure AI Engineer Associate (AI-102) | Earned Sep 2025

Selected training: RAG with LangChain (DataCamp, 2025); Advanced Android Development Nanodegree (Udacity, 2022)